

1. Write a program to check if a number is even or odd.

```
In [40]: x = int(input("Enter a number: "))
if x%2==0 :
    print(f'{x} is even')
else:
    print(f'{x} is odd')
```

5 is odd

2. Check if a person is eligible to vote (age 18 or above).

```
In [41]: a = int(input("Enter the age: "))
if a >= 18:
    print('Person is eligible to vote')
else:
    print("Person is not eligible to vote")
```

Person is eligible to vote

3. Determine if a given year is a leap year or not.

```
In [42]: a = int(input('Enter the year to check if it is leap year or not: '))
if a % 4 == 0:
    print(f'{a} is a leap year')
else:
    print(f'{a} is not a leap year')
```

2000 is a leap year

4. Check if a number is positive, negative, or zero.

```
In [43]: x = int(input("Enter a number"))
if x > 0:
    print('Positive number')
elif x == 0:
    print(' zero ')
elif x<0:
    print("is negative number")
else:
    print("Enter a valid input")
```

Positive number

5. Write a program to find the greatest of two numbers.

```
In [44]: a = int(input('enter first number:'))
b = int(input('enter second number: '))
if a > b:
    print(f'{a} is greater than {b}')
elif b>a:
    print(f'{b} is greater than {a}')
elif a ==b:
    print(f'{a} and {b} are equal')
```

5 and 5 are equal

6. Determine if a number is a multiple of 5

```
In [45]: x = int(input('enter a number:'))
if x % 5 == 0:
    print(f'{x} is a multiple of 5')
else:
    print(f'{x} is not a multiple of 5')
```

5 is a multiple of 5

7. Check if a character is a vowel or consonant.

```
In [46]: s = input('Enter a character:').lower()
if s in ('aeiou'):
    print(f'{s} is a vowel')
else:
    print(f'{s} is a consonant')
```

5 is a consonant

8. Determine if a person is eligible for a senior citizen discount (age 60+)

```
In [47]: age = int(input('Enter your age: '))
if age >= 60:
    print('You are eligible for senior discount')
else:
    print('You are not eligible for senior discount')
```

You are not eligible for senior discount

9. Write a program to check if a number is a single-digit number.

```
In [48]: number = int(input('Enter a number: '))
if number < 10 and number > -10:
    print(f'{number} is a single digit number')
else:
    print(f'{number} is not a single digit number')
```

5 is a single digit number

10. Print "Good Morning" if the time is before 12 PM, otherwise print "Good Afternoon".

```
In [49]: time = int(input('Enter time in 24-hour format (0-23): '))
if time < 12:
    print('Good Morning')
elif time < 18:
    print('Good Afternoon')
else:
    print('Good Evening')
```

Good Morning

11. Check if a string is empty or not.

```
In [50]: s = input('Enter a string: ')
if not s:
    print('String is empty')
else:
    print('String is not empty')
```

String is not empty

12. Verify if a number is a perfect square.

```
In [51]: import math as m

num = int(input('Enter a number: '))
if m.sqrt(num)**2==num:
    print(f'{num} is a perfect square')
else:
    print(f'{num} is not a perfect square')
```

5 is not a perfect square

13. Determine if a number is between 1 and 100

```
In [52]: num = int(input('Enter a number: '))
if num >0 and num <=100:
    print(f'{num} is in the range of 1 to 100')
else:
    print(f'{num} is not in the range of 1 to 100')
```

5 is in the range of 1 to 100

14. Print "Weekend" if the day is Saturday or Sunday; otherwise, print "Weekday".

```
In [53]: day = input('Enter a day of the week: ').lower()
if day in('saturday', 'sunday'):
    print(f'{day} is a weekend')
else:
    print(f'{day} is a weekday')
```

5 is a weekday

15. Find if a given number is exactly divisible by both 3 and 7.

```
In [54]: num = int(input('Enter a number: '))
if num%3==0 and num%7==0:
    print(f'{num} is divisible by both 3 and 7')
elif num%3==0:
    print(f'{num} is divisible by 3')
elif num%7==0:
    print(f'{num} is divisible by 7')
else:
    print(f'{num} is not divisible by 3 or 7')
```

5 is not divisible by 3 or 7

16. Check if the sum of two numbers is greater than 100

```
In [55]: num1 = int(input('Enter a number: '))
num2 = int(input('Enter another number: '))
add = num1 + num2
if add > 100:
    print(f'The sum of {num1} and {num2} is {add}, which is greater than 100')
else:
    print(f'The sum of {num1} and {num2} is {add}, which is not greater than 100')
```

The sum of 5 and 5 is 10, which is not greater than 100

17. Write a program to find the minimum of two numbers.

```
In [56]: num1 = int(input('Enter a number: '))
num2 = int(input('Enter another number: '))
if num1 > num2:
    print(f'{num1} is greater than {num2}')
else:
    print(f'{num2} is greater than {num1}')
```

5 is greater than 5

18. Check if a number is divisible by 2 but not by 3

```
In [57]: num = int(input('Enter a number: '))
if num%2==0 and num%3!=0:
    print(f'{num} is divisible by both 2 and 3')
elif num%2==0:
    print(f'{num} is divisible by 2 and not by 3')
elif num%3==0:
    print(f'{num} is divisible by 3 and not by 2')
else:
    print(f'{num} is not divisible by either 2 or 3')
```

5 is not divisible by either 2 or 3

19. Determine if a given alphabet is uppercase or lowercase.

```
In [58]: s = input('enter an alphabet: ')
if s.isupper():
    print(f'{s} is a uppercase letter')
else:
    print(f'{s} is a lowercase letter')
```

5 is a lowercase letter

20. Check if a triangle is valid given three side lengths (sum of any two sides must be greater than the third).

```
In [59]: a=int(input('Enter 1st side of triangle:'))
b=int(input('Enter 2nd side of triangle:'))
c=int(input('Enter 3rd side of triangle:'))

if a+b>c:
    print('this is a valid triangle')
elif b+c>a:
    print('this is a valid triangle')
elif a+c>b:
    print('this is a valid triangle')
```

```
else:  
    print('this is not a valid triangle')
```

this is a valid triangle

21. Find the largest of three numbers.

```
In [60]: a = int(input('Enter a number: '))  
b = int(input('Enter another number: '))  
c = int(input('Enter a third number: '))  
if a > b and a > c:  
    print(a, 'is the greatest number')  
elif b > a and b > c:  
    print(b, 'is the greatest number')  
else:  
    print(c, 'is the greatest number')
```

5 is the greatest number

22. Determine if a number is a prime number.

```
In [61]: num = int(input("Enter a number: "))  
if num > 1:  
    for i in range(2, int(num**0.5) + 1):  
        if num % i == 0:  
            print("Not a prime number")  
            break  
    else:  
        print("Prime number")  
else:  
    print("Not a prime number")
```

23. Check if a person is eligible for a driving license (age 18+, passed the driving test).

```
In [62]: age = int(input('Enter your age: '))  
if age >= 18:  
    print('You are eligible for a driving license')  
else:  
    print('You are not eligible for a driving license')
```

You are eligible for a driving license

24. Write a program to determine whether a triangle is equilateral, isosceles, or scalene.

```
In [63]: a = int(input('Enter first side of triangle: '))  
b = int(input('Enter second side of triangle: '))  
c = int(input('Enter third side of triangle: '))  
if a == b and b == c:  
    print('The triangle is equilateral')  
elif a == b or b == c or a == c:  
    print('The triangle is isosceles')  
else:  
    print('The triangle is scalene')
```

The triangle is scalene

25. Determine if a student passes or fails based on a passing mark of 40

```
In [64]: marks = int(input('Enter your marks: '))
if marks >= 40:
    print('You have passed')
else:
    print('You have failed')
```

You have passed

26. Check if a number is a palindrome (same forward and backward)

```
In [65]: num = int(input('Enter a number: '))
if str(num) == str(num)[::-1]:
    print(f'{num} is a palindrome')
else:
    print(f'{num} is not a palindrome')
```

56 is not a palindrome

27. Calculate the electricity bill based on consumption: ₹5 per unit for the first 100 units, ₹10 per unit for the next 200, ₹15 per unit for anything above 300 units

```
In [66]: units = int(input('Enter the number of units consumed: '))
if units <= 100:
    bill = units * 5
    print(f'Your bill is ₹{bill}')
elif units <= 200:
    bill = 100*5 + (units-100)*10
    print(f'Your bill is ₹{bill}')
elif units <= 300:
    bill = 100*5 + 100*10 + (units-200)*15
    print(f'Your bill is ₹{bill}')
else:
    print('Your bill is too high to calculate')
```

Your bill is ₹280

28. Find the grade of a student based on marks (90+ A, 80-89 B, etc.)

```
In [67]: marks = int(input('Enter your marks: '))
if marks > 90:
    print('You got A grade')
elif marks >= 80 and marks < 90:
    print('You got B grade')
elif marks >= 70 and marks < 80:
    print('You got C grade')
elif marks >= 60 and marks < 70:
    print('You got D grade')
elif marks >= 50 and marks < 60:
    print('You got E grade')
else:
    print('You got F grade')
```

You got B grade

29. Determine if a given date is valid (considering month length and leap year for February)

In []:

30. Check if a given time is AM or PM

```
In [68]: time = int(input('Enter time in 24-hour format (0-23): '))
if time<12:
    print('AM')
else:
    print('PM')
```

PM

31. Check if a number is an Armstrong number

```
In [69]: num = input("Enter a number: ")
power = len(num)
if sum(int(digit) ** power for digit in num) == int(num):
    print(f"{num} is an Armstrong Number")
else:
    print(f"{num} is not an Armstrong Number")
```

5 is an Armstrong Number

32. Determine the type of quadrilateral based on given angles and sides.

```
In [70]: a = int(input('Enter first side of quadrilateral: '))
b = int(input('Enter second side of quadrilateral: '))
c = int(input('Enter third side of quadrilateral: '))
d = int(input('Enter fourth side of quadrilateral: '))
if a==b and b==c and c==d and d==a:
    print('The quadrilateral is a square')
elif a==c and b==d:
    print('The quadrilateral is a rectangle')
elif a==b or b==c or c==d or d==a:
    print('The quadrilateral is a rhombus')
else:print('Its a quadrilateral but not a square, rectangle or rhombus')
```

The quadrilateral is a rectangle

33. Implement a basic calculator that takes two numbers and an operation (+, -, *, /)

```
In [71]: a = int(input('Enter first number: '))
b = int(input('Enter second number: '))
operator = input('Enter an operator (+, -, *, /): ')
if operator == '+':
    print(f'The sum of {a} and {b} is {a + b}')
elif operator == '-':
    print(f'The difference of {a} and {b} is {a - b}')
elif operator == '*':
    print(f'The product of {a} and {b} is {a * b}')
elif operator == '/':
```

```

if b != 0:
    print(f'The division of {a} by {b} is {a / b}')
else:
    print('Division by zero is not allowed')
else:
    print('Invalid operator. Please use +, -, *, or /.')

```

The product of 5 and 5 is 25

34. Check if a bank account balance is sufficient for withdrawal

```

In [72]: balance = int(input("Enter your current balance: "))
amount = int(input("Enter the amount: "))
if amount <= balance:
    balance = balance - amount
    print(f"Transaction successful! You withdrew {amount} and your new balance is ")
else:
    print("Insufficient balance for this transaction.")

```

Transaction successful! You withdrew 14526 and your new balance is: 185474

35. Implement a simple temperature converter (Celsius to Fahrenheit and vice versa).

```

In [73]: tempe= int(input("Enter temperature: "))
a = input("Enter 'C' for Celsius to Fahrenheit or 'F' for Fahrenheit to Celsius: ")
if a == 'C':
    fahrenheit = (tempe * 9/5) + 32
    print(f"{tempe}°C is equal to {fahrenheit}°F")
elif a == 'F':
    celsius = (tempe - 32) * 5/9
    print(f"{tempe}°F is equal to {celsius}°C")
else:
    print("Invalid input. Please enter 'C' or 'F'.")

```

86°C is equal to 186.8°F

36. Write a program to check if a number lies within a specific range (e.g., between 50 and 100)

```

In [74]: num = int(input('Enter a number: '))
if 50 <= num <= 100:
    print(f'{num} is in the range of 50 to 100')
else:
    print(f'{num} is not in the range of 50 to 100')

```

55 is in the range of 50 to 100

37. Determine if a given year is a century year (ends in 00).

```

In [75]: year = int(input('Enter a year: '))
if year % 100 == 0:
    print(f'{year} is a century year')
else:
    print(f'{year} is not a century year')

```

2004 is not a century year

38. Check if a given integer is a power of 2.

```
In [76]: num = int(input('Enter a number: '))
if num > 0 and (num & (num - 1)) == 0:
    print(f"{num} is a power of 2")
else:
    print(f"{num} is not a power of 2")
```

5 is not a power of 2

40. Validate a password (must be at least 8 characters long and contain both numbers and letters)

```
In [82]: import re
password = input("enter your password: ")
if len(password) >= 8 and re.search(r'[A-Za-z]', password) and re.search(r'[0-9]', password):
    print("valid password")
else:
    print("invalid password")
```

valid password

41. Implement a ticket pricing system where children under 5 enter for free, seniors (60+) get a 50% discount, and others pay ₹100

```
In [77]: age = int(input('Enter your age: '))
if age < 5:
    print('You can enter for free')
elif age >= 60:
    price = 100 * 0.5
    print(f'You get a 50% discount on the ticket price. Your ticket price is ₹{price}')
else:
    print('You need to pay ₹100 for the ticket')
```

You need to pay ₹100 for the ticket

42. Check if three numbers can form a Pythagorean triplet.

```
In [78]: a = int(input('Enter 1st number:'))
b = int(input('Enter 2nd number:'))
c = int(input('Enter 3rd number:'))
if a**2 + b**2 == c**2 or b**2 + c**2 == a**2 or a**2 + c**2 == b**2:
    print(f'{a}, {b}, {c} can form a Pythagorean triplet')
else:
    print(f'{a}, {b}, {c} cannot form a Pythagorean triplet')
```

5, 8, 9 cannot form a Pythagorean triplet

44. Determine the zodiac sign based on a given birth date.

```
In [79]: day = int(input('Enter day of the month (1-31): '))
month = int(input('Enter month (1-12): '))
if (month == 1 and day >= 20) or (month == 2 and day <= 18):
    print('your zodiac sign is Aquarius')
elif (month == 2 and day >= 19) or (month == 3 and day <= 20):
```

```

    print('your zodiac sign is Pisces')
elif (month == 3 and day >= 21) or (month == 4 and day <= 19):
    print(' your zodiac sign isAries')
elif (month == 4 and day >= 20) or (month == 5 and day <= 20):
    print('your zodiac sign is Taurus')
elif (month == 5 and day >= 21) or (month == 6 and day <= 20):
    print('your zodiac sign is Gemini')
elif (month == 6 and day >= 21) or (month == 7 and day <= 22):
    print('your zodiac sign is Cancer')
elif (month == 7 and day >= 23) or (month == 8 and day <= 22):
    print('your zodiac sign is Leo')
elif (month == 8 and day >= 23) or (month == 9 and day <= 22):
    print('your zodiac sign is Virgo')
elif (month == 9 and day >= 23) or (month == 10 and day <= 22):
    print('your zodiac sign is Libra')
elif (month == 10 and day >= 23) or (month == 11 and day <= 21):
    print('your zodiac sign is Scorpio')
elif (month == 11 and day >= 22) or (month == 12 and day <= 21):
    print('your zodiac sign is Sagittarius')
elif (month == 12 and day >= 22) or (month == 1 and day <= 19):
    print('your zodiac sign is Capricorn')
else:
    print('Invalid date')

```

your zodiac sign is Libra

48. Implement a loan eligibility checker based on income, credit score, and employment status.

```

In [80]: income = int(input("Enter your income: "))
creditscore = int(input("Enter your credit score: "))
emp_status = input("Are you employed? (yes/no): ")
if emp_status.lower()== 'yes':
    if income>=50000 and creditscore>=650:
        print('You are eligible for loan')
    else:
        print('You are not eligible for loan')

```

You are eligible for loan

49. Implement a rock-paper-scissors game using conditional statements

```

In [81]: your_move = input("Enter your move: ")
opp_move = input("Enter opponent's move: ")
if your_move == opp_move:
    print("It's a tie!")
elif your_move == "rock" and opp_move == "scissors":
    print("You win!")
elif your_move == "paper" and opp_move == "rock":
    print("You win!")
elif your_move == "scissors" and opp_move == "paper":
    print("You win!")
else:
    print("You lose!")

```

You lose!

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